



**A
PROJECT
ON**

“Web Application of Mayur Super Market”

SUBMITTED TO

SHIVAJI UNIVERSITY, KOLHAPUR

FOR THE DEGREE OF

MASTER OF COMPUTER APPLICATION

MCA I SEM II

UNDER THE FACULTY OF

COMMERCE AND MANAGEMENT

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YSPM'S YASHODA TECHNICAL CAMPUS, SATARA

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Yashoda Shikshan Prasarak Mandal's
YASHODA TECHNICAL CAMPUS, SATARA

(An Autonomous Institute)

FACULTY OF COMPUTER APPLICATION

Department of Master of Computer Application

Affiliated to Shivaji University, Kolhapur | Approved by AICTE, DTE Mumbai

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DECLARATION

I declare that the synopsis titled **“Web Application Of Mayur Super Market”** has been prepared under the guidance of Dr, Ranika Mam and support from **Arun Bhosale**. This synopsis is submitted to the **Department of Computer Application, Yashoda Technical Campus, Satara**, as part of the requirements for the **MCA II SEM III Major Project**.

Student Sign

Project Guide Sign

MCA HOD

Index

Sr. No.	Description	Page No.
1.	Introduction • About the Project • Existing System • Need and Scope of the New System • Organization Details	
2.	Proposed System • Objective • H/W S/W Specification • Technology used • Database Design (DFD, ERD)	
3.	Working of Proposed System	
4.	Advantages	
5.	Future Scope	
6.	Conclusion	
7.	References	

Introduction

Mayur Super Market Management System is a web-based application developed to automate and simplify the daily operations of a supermarket. In today's fast-paced retail environment, supermarkets handle a large number of products, customers, suppliers, sales transactions, and inventory records every day. Managing these activities manually is time-consuming, error-prone, and inefficient. This project provides a computerized solution that helps the supermarket owner manage all business operations accurately, efficiently, and securely.

The main objective of the Mayur Super Market Management System is to improve the overall management of the supermarket by replacing manual record-keeping with a digital system. The application allows the administrator to maintain product information, categories, suppliers, customers, employees, purchases, inventory, sales, billing, and reports through a single integrated platform. It helps reduce paperwork, minimize human errors, and increase the speed and accuracy of daily business operations.

The system is designed with a user-friendly interface so that the administrator can easily perform different tasks without requiring advanced technical knowledge. The administrator can add, update, delete, and search records related to products, suppliers, customers, and employees. The inventory module automatically updates stock levels whenever products are purchased or sold, ensuring that accurate stock information is always available. This helps prevent stock shortages, overstocking, and product wastage.

One of the important features of the system is sales and billing management. During each sales transaction, the system automatically calculates the total amount, generates the bill, updates product stock, and stores the transaction details in the database. This saves time, improves billing accuracy, and enhances customer service. The purchase management module records all products purchased from suppliers, making it easier to monitor inventory replenishment and supplier performance.

The reporting module provides valuable business insights by generating reports on daily, weekly, monthly, and yearly sales, purchases, stock availability, and customer information. These reports help the supermarket owner analyze business performance, identify best-selling products, monitor inventory levels, and make informed business decisions. Proper reporting also improves transparency and simplifies business management.

The system ensures data security by providing an administrator login, allowing only authorized users to access and manage the application. This protects confidential business information and prevents unauthorized modifications to the database.

The Mayur Super Market Management System is developed using modern web technologies. The frontend is designed using **React.js, HTML, CSS, and JavaScript** to provide a responsive and attractive user interface. The backend is developed using **Python Django**, while **MySQL** is used as the database management system for storing and retrieving information efficiently. The application follows a modular architecture, making it easy to

maintain, update, and expand with new features in the future. These technologies are commonly used for developing secure and scalable business applications.

Overall, the Mayur Super Market Management System offers a complete solution for managing supermarket operations efficiently. It improves productivity, reduces manual work, enhances data accuracy, speeds up billing, simplifies inventory management, and supports better business decision-making through detailed reports. The system is designed to provide a reliable, secure, and efficient management platform that meets the day-to-day operational requirements of a modern supermarket and contributes to its growth and customer satisfaction.

Existing System

The existing system used in many small and medium-sized supermarkets is primarily manual or partially computerized. Most daily operations such as maintaining product records, inventory management, sales transactions, supplier details, customer information, and employee records are handled using paper registers, notebooks, spreadsheets, or separate software applications. This traditional approach is time-consuming, difficult to manage, and often results in data duplication and human errors.

In the manual system, all product details such as product name, category, price, quantity, and stock availability are recorded in registers or spreadsheets. Whenever new stock arrives, employees manually update the records, and during sales, stock quantities are reduced manually. Since these updates depend on human effort, there is a high possibility of incorrect stock calculations, duplicate entries, and missing records. As a result, the supermarket may experience stock shortages, overstocking, or inaccurate inventory information.

Sales and billing are also handled manually or with basic billing software that is not integrated with inventory management. Cashiers prepare bills, calculate totals, and record transactions separately. Stock information is not updated automatically after each sale, requiring employees to update inventory records later. This process increases the chances of calculation errors, delays, and inconsistencies between sales records and inventory.

Supplier and purchase management is another challenge in the existing system. Information about suppliers, purchase orders, and delivered products is stored in paper files or different registers. Tracking previous purchases, comparing supplier performance, or checking outstanding orders becomes difficult. Searching for specific records consumes significant time and effort.

Customer information is generally maintained only for regular customers or is not stored at all. Without a proper customer database, supermarkets cannot analyze customer purchasing behavior, maintain purchase history, or improve customer relationship management.

Employee management is also performed manually. Attendance, salary records, job roles, and contact details are maintained in separate registers or spreadsheets. Updating employee

information and preparing reports require additional manual work, increasing administrative effort and the possibility of errors.

Generating business reports is one of the biggest limitations of the existing system. Daily sales reports, purchase summaries, inventory reports, and financial records are prepared manually, which takes considerable time and may not always be accurate. Since information is scattered across different files and registers, management cannot quickly access real-time business data for decision-making.

The existing system also lacks proper security. Paper records can be lost, damaged, or accessed by unauthorized individuals. Spreadsheet-based systems may also be modified accidentally without maintaining proper backup or user authentication. This creates risks related to data integrity, confidentiality, and business continuity.

Another major drawback is the absence of integration between different business functions. Product management, inventory, sales, purchases, suppliers, employees, and customer records are maintained separately, making it difficult to coordinate information across departments. As the supermarket grows, the manual system becomes increasingly inefficient and difficult to maintain.

Limitations of the Existing System

- Manual data entry increases the chances of human error.
- Paper-based record keeping requires excessive time and effort.
- Inventory is not updated automatically after sales or purchases.
- Billing process is slower and prone to calculation mistakes.
- Difficulty in searching and retrieving old records.
- Poor management of supplier, customer, and employee information.
- Reports are generated manually and require significant time.
- No centralized database for storing business information.
- Lack of proper data security and backup facilities.
- Increased paperwork and storage requirements.
- Difficult to monitor stock availability in real time.
- Limited support for business analysis and decision-making.
- Not suitable for handling large volumes of supermarket transactions.

Due to these limitations, there is a need for an integrated, secure, and automated Super Market Management System that can efficiently manage all supermarket operations from a single platform. The proposed system addresses these issues by providing centralized data management, automatic inventory updates, secure user authentication, fast billing, and comprehensive reporting facilities.

Need and Scope of the System

Need of the System

In today's competitive retail industry, supermarkets deal with a large number of products, customers, suppliers, employees, and sales transactions every day. Managing all these activities manually is difficult, time-consuming, and prone to errors. Traditional paper-based record keeping or maintaining separate spreadsheets creates problems such as duplicate entries, inaccurate inventory records, delayed billing, and difficulty in generating reports.

The **Mayur Super Market Management System** is developed to overcome these challenges by providing an integrated and automated solution for managing supermarket operations. The system reduces manual work, improves operational efficiency, and ensures that all business information is stored securely in a centralized database.

The need for this system arises because it helps supermarket owners and administrators perform daily business activities quickly and accurately. It minimizes human errors, saves time, improves inventory control, and enables faster decision-making through real-time reports. The system also provides secure access through administrator authentication, protecting sensitive business data from unauthorized users.

The application automatically updates stock whenever products are purchased or sold, helping prevent stock shortages and overstocking. It also simplifies billing, purchase management, supplier management, customer record maintenance, and employee management. Overall, the system enhances productivity, improves customer service, and supports the smooth functioning of supermarket operations.

Need of the System

- To replace manual record-keeping with a computerized system.
- To maintain product, customer, supplier, and employee records efficiently.
- To automate inventory management and stock updates.
- To reduce paperwork and human errors.
- To generate accurate sales, purchase, and inventory reports.
- To improve billing speed and transaction accuracy.
- To provide secure storage of business information.
- To save time and reduce operational costs.
- To improve overall supermarket management and productivity.
- To support better business planning and decision-making.

Scope of the System

The **Mayur Super Market Management System** provides a complete solution for managing the day-to-day operations of a supermarket through a single web-based platform. The system covers all major business activities, including product management, inventory control, sales, purchases, supplier management, customer management, employee management, billing, and report generation.

The application is designed primarily for the supermarket administrator, who has the authority to manage all records and monitor overall business performance. The administrator can add, update, delete, and search information related to products, categories, suppliers, customers, employees, purchases, and sales. The system automatically maintains inventory levels after every purchase and sales transaction, ensuring accurate stock information.

The reporting module allows the administrator to generate daily, weekly, monthly, and yearly reports related to sales, purchases, inventory, and customer transactions. These reports help management analyze business performance, monitor stock levels, and make informed business decisions.

The system is developed using **React.js** for the frontend, **Python Django** for the backend, and **MySQL** as the database management system. It provides a secure login mechanism for administrators and stores all business information in a centralized database.

Scope of the System

- Product and category management.
- Inventory and stock management.
- Supplier and purchase management.
- Customer information management.
- Employee management.
- Sales and billing management.
- Automatic stock updates after purchases and sales.
- Search and filter functionality for records.
- Report generation for sales, purchases, inventory, and customers.
- Secure administrator login and authentication.
- Centralized database management.
- Fast, accurate, and reliable business operations.
- Easy maintenance and future system expansion.

The scope of the project is limited to the internal management of supermarket operations. It focuses on improving administrative efficiency by automating routine business processes and providing accurate information for effective decision-making. The modular design also allows future enhancements, such as barcode scanning, online ordering, digital payment integration, SMS or email notifications, and advanced business analytics.

Organization Details

Client : To reduce manual work, save time, maintain accurate records, and easily access supermarket information through one system.

2. Proposed System

Objectives of the System

1. To develop a web-based supermarket management system that automates daily business operations.
2. To maintain product, category, supplier, customer, and employee information in a centralized database.
3. To automate inventory management by updating stock automatically after every purchase and sale.
4. To provide a fast and accurate sales and billing system that reduces customer waiting time.
5. To simplify purchase management by maintaining supplier details and purchase records.
6. To reduce paperwork, duplicate records, and human errors through automation.
7. To provide secure administrator login and authentication for protecting business data.
8. To generate accurate sales, purchase, inventory, and employee reports for effective decision-making.
9. To improve the speed, efficiency, and reliability of supermarket operations.
10. To provide quick search, update, and deletion of records through an easy-to-use interface.
11. To enhance inventory control by monitoring stock availability and reducing stock shortages or overstocking.
12. To improve overall business productivity by integrating all supermarket activities into a single system.
13. To ensure data consistency, accuracy, and secure storage of business information.
14. To create a scalable system that can be extended with future features such as barcode scanning, online ordering, digital payment integration, and advanced business analytics.

H/W & S/W Specification

Hardware Specification

- **Processor:** Intel Core i3 or higher
- **RAM:** 4 GB
- **Storage:** 256 GB SSD
- **Monitor:** LED Monitor
- **Keyboard:** Standard Keyboard
- **Mouse:** Optical Mouse

Software Specification

- **Operating System:** Windows 11
- **Frontend:** React.js, HTML5, CSS3, JavaScript
- **Backend:** Python Django
- **Database:** MySQL
- **Code Editor:** Visual Studio Code
- **Web Browser:** Google Chrome
- **Version Control:** Git & GitHub

Technology Used

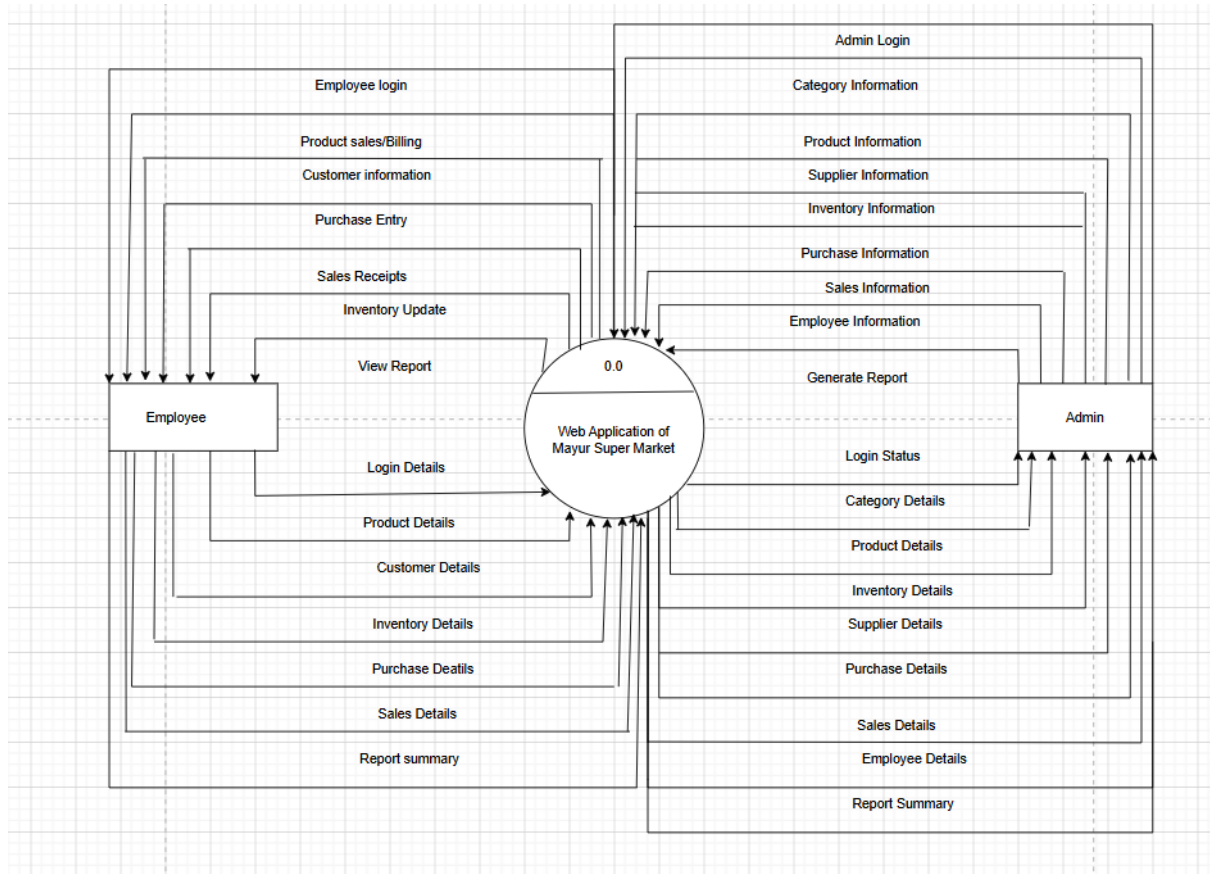
The following technologies are used in the development of the **Mayur Super Market Management System**:

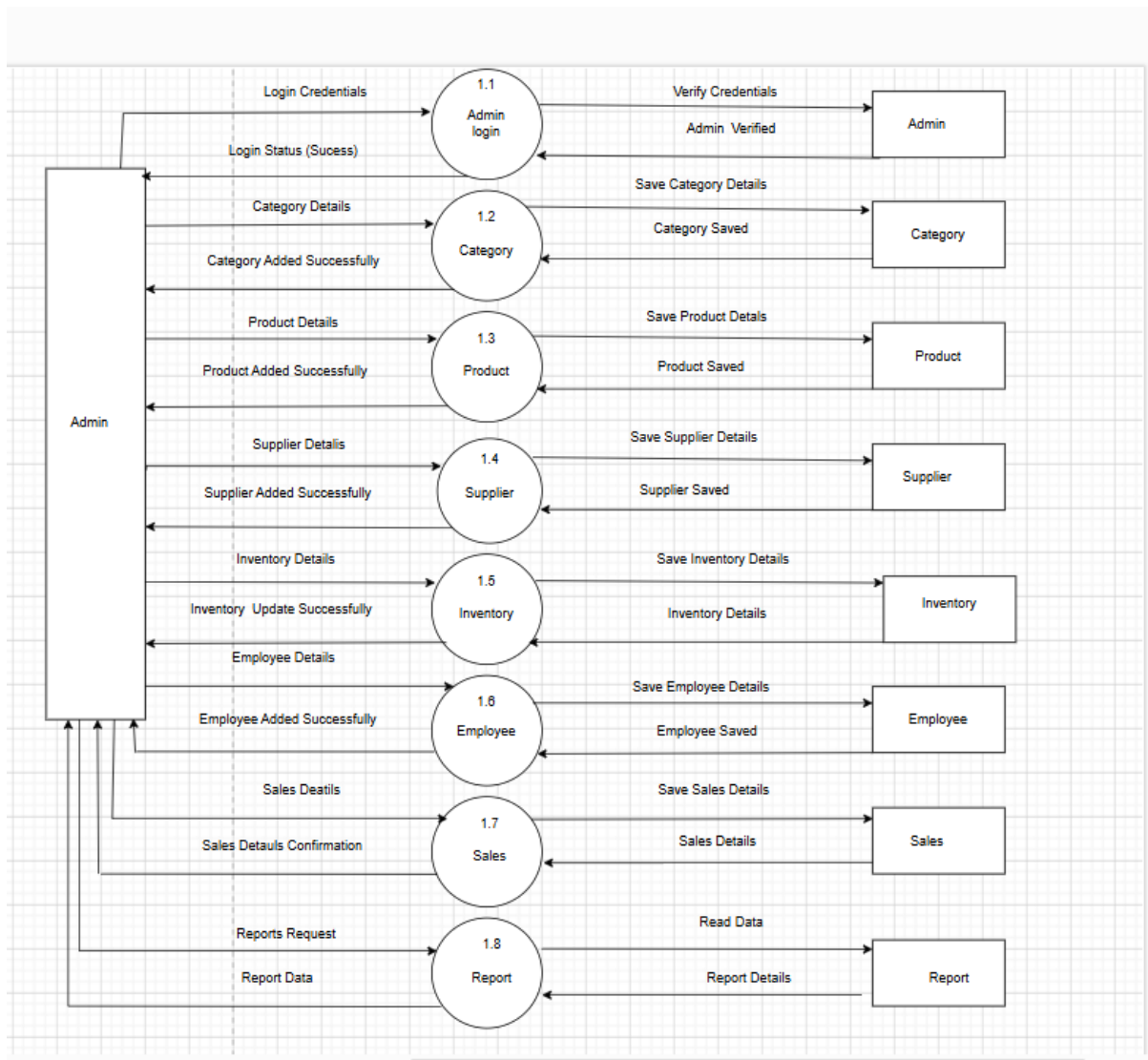
- **Frontend:** React.js, HTML5, CSS3, JavaScript
- **Backend:** Python Django
- **Database:** MySQL
- **Programming Language:** Python
- **Code Editor:** Visual Studio Code
- **Web Browser:** Google Chrome
- **Version Control:** Git & GitHub
- **Operating System:** Windows 11

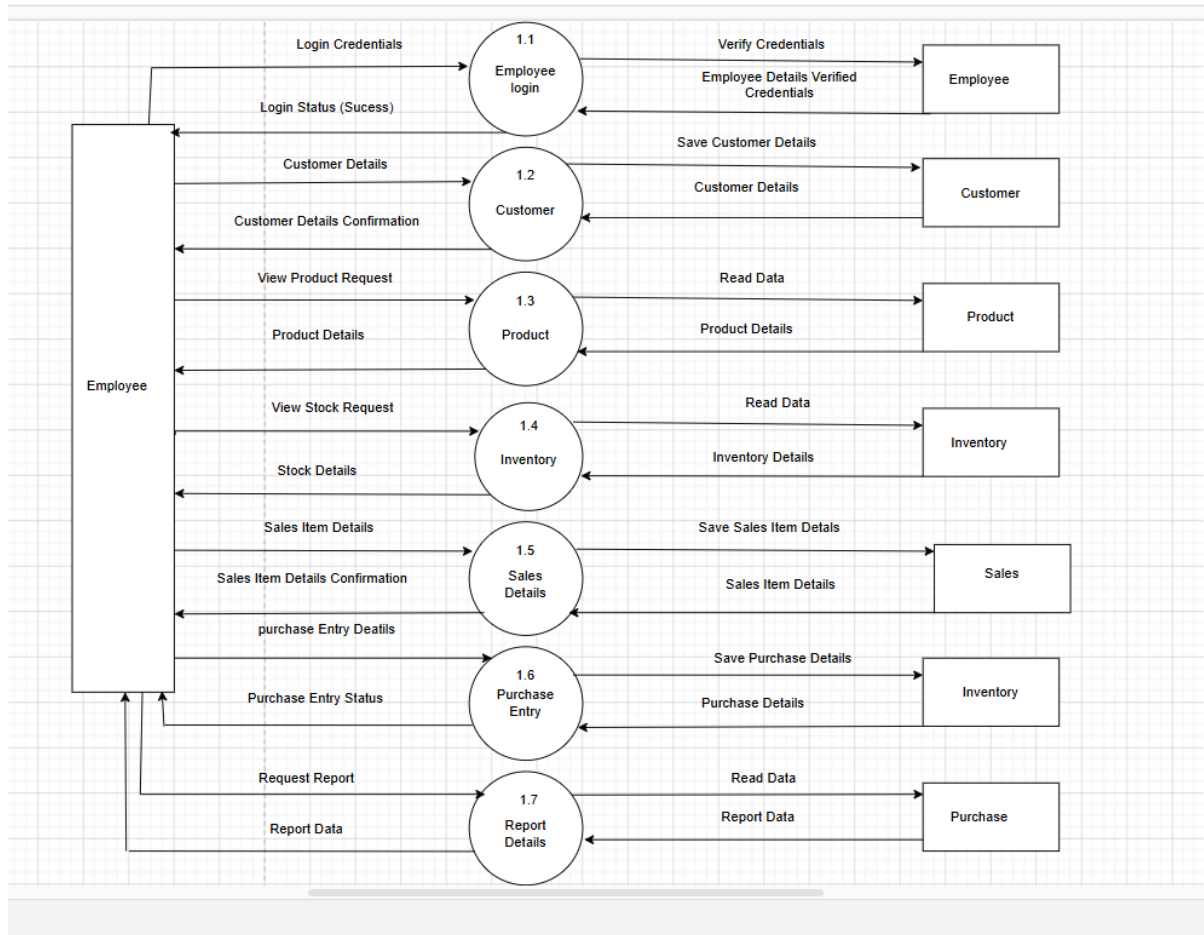
System Diagram

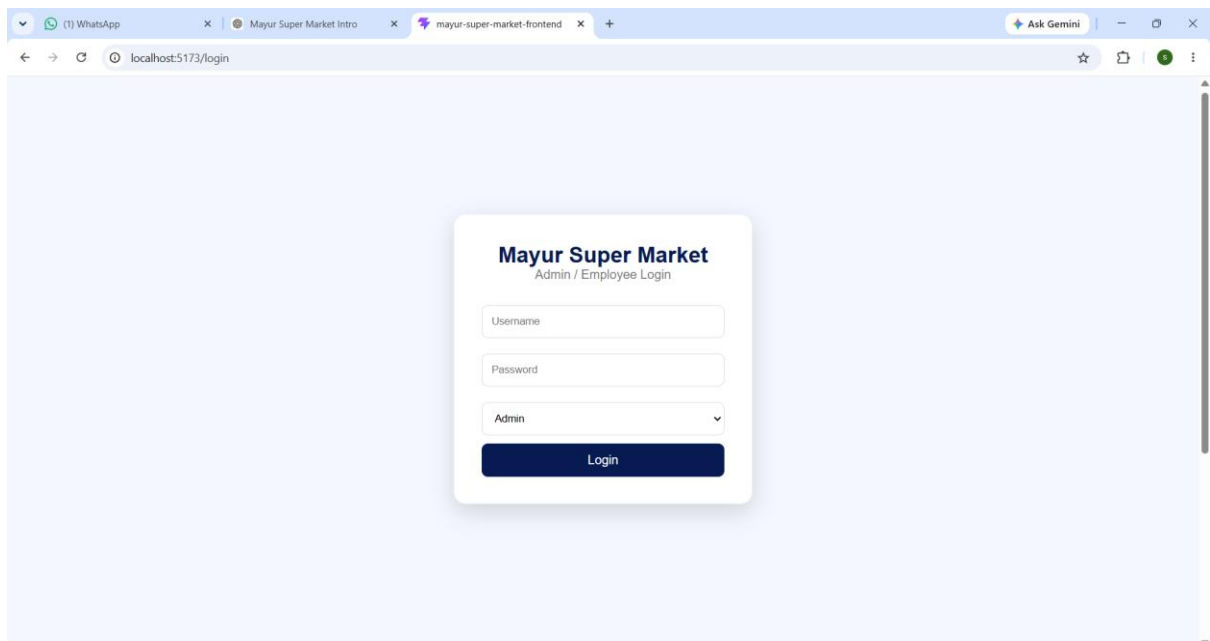
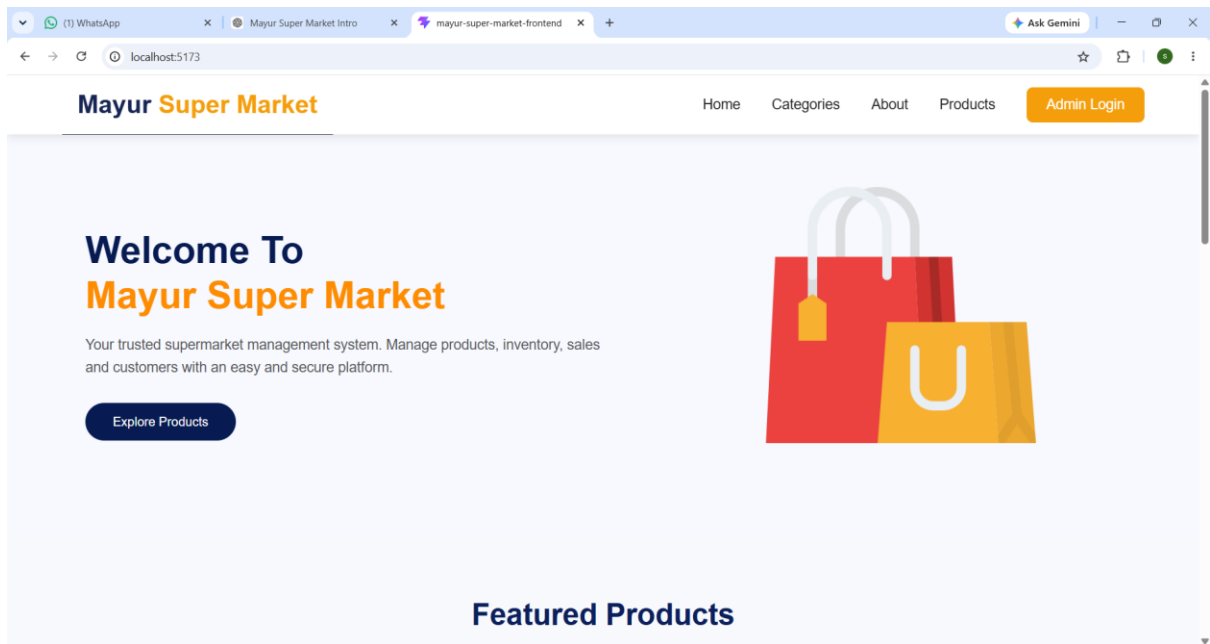
DFD (Data Flow Diagram)

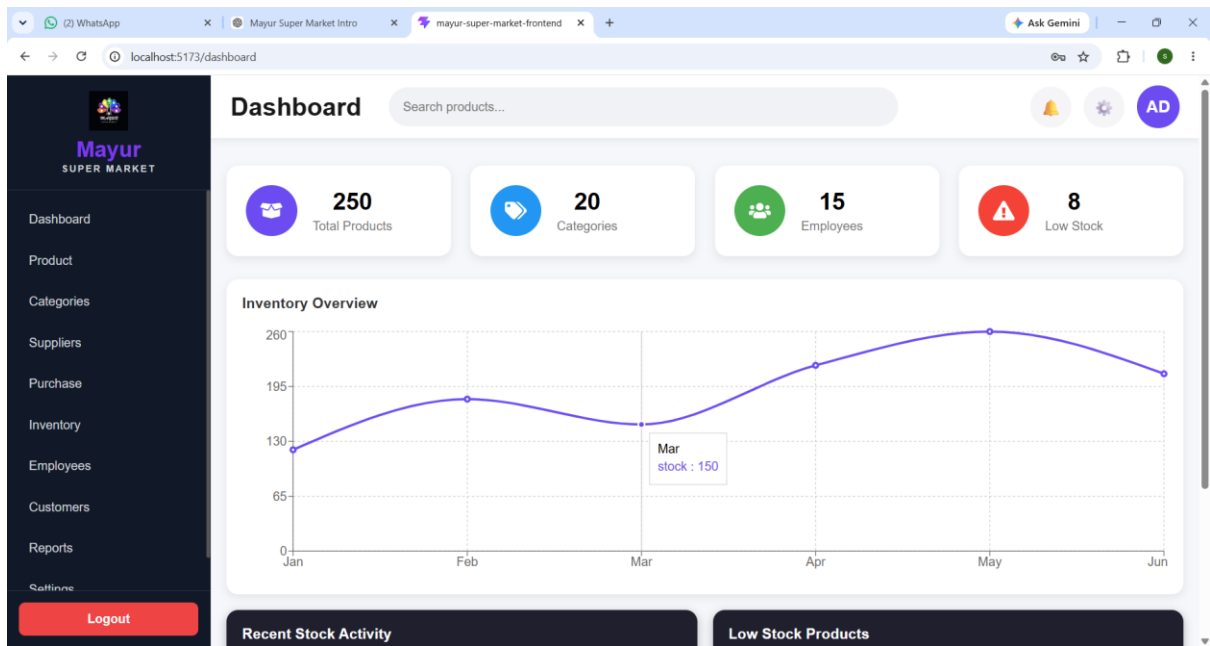
Zero Level D











WhatsApp Mayur Super Market Intro mayur-super-market-frontend Ask Gemini localhost:5173/productD

Dashboard

Search products...

Products

Product Name

Category

Price

Stock

Add Product

Search Product...

ID	Product Name	Category	Price	Stock	Status	Action
1	Rice	Groceries	₹60	120	In Stock	EditDelete
2	Sugar	Groceries	₹45	80	In Stock	EditDelete
3	Milk	Dairy	₹30	50	In Stock	EditDelete

4.Advantages

1. Saves time.
2. Reduces manual work.
3. Keeps data safe and organized.
4. Makes stock tracking easy.
5. Reduces errors in records.
6. Makes product and customer records easy to manage.
7. Helps manage purchases and sales.
8. Makes report generation easy.
9. Provides separate access for Admin and Employee.
10. Makes supermarket work simple and efficient.

5.Future Scope of Mayur Super Market

The **Mayur Super Market** system can be improved with more features in the future. These features can make the system faster, more useful, and easier to use.

1. Mobile Application

A mobile application can be developed in the future so that the Admin and Employees can access the system easily from their mobile phones.

2. Barcode Scanning

Barcode scanning can be added to quickly identify products, update stock, and reduce data entry errors.

3. Low Stock Notifications

The system can send notifications when the quantity of a product becomes low. This will help the Admin purchase products on time.

4. Online Backup

Data can be stored safely using cloud or online backup. This can help prevent data loss.

5. Advanced Reports

More detailed reports and charts can be added for products, purchases, sales, stock, and overall store activities.

